IIT Gandhinagar

Sleep Apnea Detection

Using Physiological Signals and Deep Learning

Technical Report — SRIP 2026

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Program	SRIP 2026 — Summer Research Internship Program
Host Institute	IIT Gandhinagar
Dataset	5 participants • ~8 hour overnight recordings • 3 physiological signals
Year	2026

Signal Processing • Digital Filtering • 1D CNN • Leave-One-Out Cross Validation • Sleep Stage Analysis

1. Introduction

Sleep apnea is a serious sleep disorder in which breathing repeatedly stops and starts during sleep. Left untreated, it leads to cardiovascular disease, hypertension, and chronic daytime fatigue. Manual identification of apnea events requires trained clinicians to annotate polysomnography (PSG) recordings, which is expensive and time-consuming at scale.

This project builds an automated end-to-end pipeline to detect two types of breathing irregularities: Hypopnea (partial airflow reduction) and Obstructive Apnea (complete airway blockage). The pipeline covers signal visualization, preprocessing, windowed feature extraction, deep learning classification, and clinical analysis by sleep stage.

Objectives:

- Visualize 8-hour physiological recordings with annotated event overlays

- Apply bandpass filtering to isolate the breathing frequency range (0.17 to 0.40 Hz)
- Segment signals into labeled 30-second windows with 50% overlap
- Train a 1D CNN using Leave-One-Participant-Out cross-validation
- Analyze apnea event distribution across sleep stages

2. Dataset Description

Overnight sleep data was collected from 5 participants (AP01 to AP05), each recorded for approximately 7.5 to 8 hours. AP01 had 875,184 nasal flow samples at 32 Hz covering 7.6 hours. Three signals were recorded per participant along with breathing event annotations and 30-second sleep stage profiles.

2.1 Physiological Signals

Signal	File Keyword	Sampling Rate	Samples (AP01)
Nasal Airflow	Flow	32 Hz	875,184
Thoracic Movement	Thorac	32 Hz	875,184
SpO2 (Oxygen Saturation)	SPO2	4 Hz	109,399

2.2 Raw File Format

```
Start Time: 5/30/2024 8:59:00 PM | Sample Rate: 32 | Data:
30.05.2024 20:59:00,000; 120
```

Timestamps use European format (DD.MM.YYYY) with comma as millisecond separator. File naming differs across participants: AP01 uses "Flow - 30-05-2024.txt", AP02 uses "Flow - 30.05.2024.txt" (double space, dots in date), AP04 uses "Flow Signal - 29.05.2024.txt". A flexible keyword-based finder handles all.

2.3 Dataset After Windowing

Label	Windows	Percentage	Description
Normal	8,038	91.3%	No event overlap exceeds 50%
Hypopnea	593	6.7%	Partial airflow reduction
Obstructive Apnea	164	1.9%	Complete airway blockage
Body event / Mixed Apnea	5	0.1%	Merged into Obstructive Apnea
TOTAL	8,800	100%	All 5 participants combined

3. Methodology

3.1 Signal Processing Pipeline

Step	Operation	Detail
1	File Parsing	Skip metadata header, detect "Data:" marker, parse DD.MM.YYYY HH:MM:SS,mmr
2	Alignment	merge_asof aligns SpO2 (4 Hz) to Flow (32 Hz) by nearest timestamp
3	Filtering	Butterworth order=4, passband 0.17-0.40 Hz, output="sos"
4	Windowing	30s windows, 50% overlap: 960 samples/window, stride=480
5	Labeling	>50% window overlap with event = event label, else Normal

3.2 Bandpass Filtering

Breathing occurs at 0.17-0.40 Hz. Cutoffs must be normalized to Nyquist (fs/2=16 Hz) before passing to butter(). Passing raw Hz values is a silent bug — filter runs without error but with completely wrong cutoffs.

```
low=0.17/16=0.0106 | high=0.40/16=0.025 |
butter(4,[low,high],btype="bandpass",output="sos")
```

3.3 Timestamp Alignment

Index 500 in Flow = t=15.6s but index 500 in SpO2 = t=125.0s. Index-based alignment is wrong. pandas merge_asof matches each Flow timestamp to the nearest SpO2 reading producing a unified 32 Hz DataFrame.

3.4 Window Labeling

```
overlap = min(event_end, window_end) - max(event_start, window_start)

If overlap / 30s > 0.5 → event label, else Normal
```

4. Model Architecture

4.1 1D Convolutional Neural Network (cnn_model.py)

The kernel slides along the time axis learning local temporal patterns across all 3 signals. PyTorch Conv1d expects (batch, channels, time) so the window is transposed from (960, 3) to (3, 960) via permute(0,2,1).

Layer	Input	Output	Notes
Conv1d(3→32,k=7)+ReLU	(B,3,960)	(B,32,960)	padding=3
MaxPool1d(2)	(B,32,960)	(B,32,480)	halves length
Conv1d(32→64,k=5)+ReLU	(B,32,480)	(B,64,480)	padding=2
MaxPool1d(2)	(B,64,480)	(B,64,240)	halves length
Conv1d(64→128,k=3)+ReLU	(B,64,240)	(B,128,240)	padding=1
AdaptiveAvgPool1d(1)	(B,128,240)	(B,128,1)	collapses time
Flatten → Linear(128→64) → Dropout → Lin	ear(64(B,128)→3)	(B,3)	classifier

4.2 Class Imbalance Handling

Baseline model predicted only Normal (Accuracy=91.1%, Recall=0.333, zero apnea detected). Two fixes applied: WeightedRandomSampler (minority windows picked more often per batch) and Weighted CrossEntropyLoss (missing Apnea penalized proportionally more than Normal).

5. Evaluation and Results

5.1 Leave-One-Participant-Out Cross Validation

Random splits cause data leakage — model memorizes person-specific breathing patterns instead of general apnea patterns. LOPO ensures no participant appears in both train and test.

Fold	Train	Test	Train Windows	Test Windows
1	AP02,AP03,AP04,AP05	AP01	6,978	1,822
2	AP01,AP03,AP04,AP05	AP02	7,031	1,769
3	AP01,AP02,AP04,AP05	AP03	7,104	1,696
4	AP01,AP02,AP03,AP05	AP04	6,868	1,932
5	AP01,AP02,AP03,AP04	AP05	7,219	1,581

5.2 Baseline 1D CNN Results

Test Participant	Accuracy	Precision (macro)	Recall (macro)
AP01	0.948	0.316	0.333

AP02	0.914	0.305	0.333
AP03	0.990	0.330	0.333
AP04	0.913	0.304	0.333
AP05	0.793	0.264	0.333
MEAN	0.911	0.304	0.333

Recall=0.333 across all folds = exactly 1/3 classes predicted (Normal only). High accuracy is misleading — model never detected Hypopnea or Apnea.

6. Sleep Stage Analysis

Sleep Profile records one stage every 30 seconds, perfectly matching window size. Five stages found across all participants: Wake, N1, N2, N3, REM.

Stage	Windows	Hypopnea	Obst. Apnea	Event Rate
Wake	3,271	10	1	0.3%
N1 (Light)	1,320	123	31	11.7%
N2 (Core)	2,440	287	72	14.7%
N3 (Deep)	1,066	60	1	5.7%
REM	650	113	59	26.5%
TOTAL	8,800	593	164	8.6%

- N2 has highest raw count (287 Hypopnea, 72 Apnea) — muscle tone decreases in core sleep
- REM has highest event rate (26.5%) — maximum muscle relaxation triggers Obstructive Apnea
- N3 deep sleep has almost zero Apnea (1 event total) — most stable breathing stage
- Wake has almost no events — patient is conscious and breathing normally

7. Challenges and Solutions

7.1 Inconsistent file naming across participants

Problem: AP01: "Flow - 30-05-2024.txt", AP02: "Flow - 30.05.2024.txt" (double space + dots), AP04: "Flow Signal", AP05: "Flow Nasal".

Solution: Flexible keyword finder strips all spaces before matching. exclude parameter prevents "Flow Events" from matching when searching for the Flow signal.

7.2 Different sampling rates between signals

Problem: Flow (32 Hz) has 875,184 samples; SpO2 (4 Hz) has 109,399. Index 500 in Flow = t=15.6s but index 500 in SpO2 = t=125.0s.

Solution: pandas merge_asof aligns by actual timestamps. Higher-frequency signal on left. Result: unified 32 Hz DataFrame with 875,184 rows.

7.3 Custom metadata header in every file

Problem: Each file has 5-6 header rows (Signal Type, Start Time, Sample Rate, Length, Unit, Data:) before numeric data begins.

Solution: Parser reads line by line, sets data_started=True when "Data:" marker is found, skips all lines before it.

7.4 Midnight crossing in event timestamps

Problem: Recording starts at 8:59 PM. Events after midnight have end_dt < start_dt when compared numerically (e.g. start=23:59, end=00:01).

Solution: After parsing, if end_dt < start_dt, add timedelta(days=1) to end_dt to correctly represent the next calendar day.

7.5 91.3% class imbalance

Problem: Baseline CNN predicted only Normal for every window across all 5 folds. Accuracy=91.1%, Recall=0.333, zero apnea events detected.

Solution: WeightedRandomSampler (weight = 1/class_count per sample) + Weighted CrossEntropyLoss with inverse frequency weights.

7.6 SpO2 filter normalization differs from 32 Hz signals

Problem: SpO2 Nyquist = 4/2 = 2 Hz. Normalized cutoffs must be 0.085 and 0.20, not 0.0106 and 0.025 used for Flow/Thorac.

Solution: bandpass_filter() takes fs as parameter and computes normalization independently per signal.

8. Conclusion and Future Work

This project built a complete end-to-end pipeline for automated sleep apnea detection handling real-world challenges including inconsistent file formats, multi-rate signal alignment, and severe class imbalance.

Completed:

- Signal visualization: 8-hour plots with Hypopnea (orange) and Apnea (red) overlays saved as PDF
- Butterworth bandpass filtering (0.17 to 0.40 Hz) isolating breathing frequency range
- Timestamp-based SpO2 alignment using merge_asof
- 8,800 labeled 30-second windows across 5 participants
- Sleep stage dataset: each window matched to Wake/N1/N2/N3/REM
- 1D CNN and ConvLSTM in PyTorch with LOPO cross-validation
- Clinical finding: N2 and REM stages have highest apnea event rates

Future Work:

- Square-root smoothing for weighted sampler to balance accuracy and recall
- Stage-conditioned models trained separately for N2 and REM
- Explicit SpO2 desaturation features as apnea biomarkers
- ConvLSTM vs 1D CNN comparison